SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

3)

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1. Introduction

A small classroom computer laboratory requires a network that allows students and the teacher to share files, access common resources, and use the internet at a low cost. Since the laboratory has limited traffic and is supervised, an economical topology is appropriate. The proposed design uses a bus-oriented network arrangement, in which workstations share a common communication path.

In the traditional implementation, all computers connect to a single RG-58 coaxial backbone. Each workstation uses a T-connector and BNC connector, while 50-ohm terminators are installed at both ends of the cable. The terminators prevent signal reflection and ensure that electrical signals do not return along the cable. A modern low-cost alternative can use Cat5e or Cat6 cabling with an inexpensive unmanaged switch to provide similar simplicity.

2. Problem Analysis

Table 1. Classroom Network Requirements

Requirement	Analysis
Users and devices	Students' computers and a teacher's computer are the primary devices.
Connectivity	All workstations communicate through a shared network medium.
Resource sharing	The teacher's PC may provide shared files using Windows or Linux file-sharing features.
Internet access	Internet access may be provided through an upstream router or gateway if required.
Cost	The design minimizes networking devices and cabling.
Reliability	The design is acceptable for light classroom usage, although a backbone failure affects all computers.
Scalability	A limited number of additional workstations may be added, subject to cable length and connection points.
Management	The network can be supervised by the teacher or laboratory administrator.

3. Proposed Network Solution

The proposed solution is a low-cost bus topology for a small classroom laboratory. A single shared backbone cable runs along the length of the classroom, and each computer connects to that backbone. The classic implementation uses RG-58, 50-ohm coaxial cable with T-connectors, BNC connectors, and 50-ohm terminators at both ends.

For a contemporary implementation, an inexpensive unmanaged switch may be used with Cat5e UTP patch cables. This option is easier to obtain and maintain than a traditional coaxial bus while preserving the low-cost objective. The choice depends on the equipment already available in the laboratory.

4. Proposed Network Architecture

```
[50-ohm Terminator]──RG-58 Shared Bus──[50-ohm Terminator]
          |             |           |
        T/BNC         T/BNC       T/BNC
          |             |           |
     Student PC 1  Student PC 2  Teacher PC
                            |
                  Shared files/resources
```

Figure 1. Proposed Bus-Topology Network Architecture

5. Network Devices and Components

Table 2. Network Devices and Components

Device or Component	Quantity	Function/Purpose
Student workstation	As required	Provides network access for each student.
Teacher’s PC	1	Provides supervision and may share files.
RG-58 coaxial cable	1 backbone run	Acts as the shared communication medium.
T-connector	One per workstation	Connects each workstation to the coaxial backbone.
BNC connector	As required	Provides the physical coaxial connection.
50-ohm terminator	2	Absorbs signals at both ends and prevents reflection.
Network interface adapter	One per computer	Connects each computer to the selected medium.
Optional unmanaged switch	1	Provides inexpensive modern Ethernet connectivity.
Optional Cat5e/Cat6 patch cables	One per workstation	Connect workstations to the switch.

6. Transmission Media and Cabling

The primary medium is RG-58 coaxial cable with 50-ohm impedance. It runs as one continuous backbone through the classroom. Each computer connects through a T-connector and BNC connector, and a 50-ohm terminator is placed at each end.

The modern alternative uses Cat5e or Cat6 UTP cable with an inexpensive unmanaged switch. Cat5e/Cat6 cabling is widely available and easier to replace than coaxial bus cabling. The exact cable length, number of workstations, and connector quantities are not specified and should be recorded as [TO BE FILLED].

7. Working Principle

When a workstation transmits data, the signal travels through the shared bus. All connected computers can detect the transmission, but only the computer whose address matches the destination processes the data. Two computers may transmit simultaneously, causing a collision; they must then wait and retransmit according to the network-access method. The 50-ohm terminators absorb the signal at both ends and prevent reflection. In the modern variation, the workstation sends its Ethernet frame to the unmanaged switch, which forwards it toward the intended device.

Student PC → T-connector/BNC → Shared RG-58 Bus → Destination PC
Teacher PC → Shared Bus → Student PCs (file/resource sharing)

8. Fault Isolation

Normal: [Terminator]──PC1──PC2──Teacher PC──PC3──[Terminator]
Workstation: [Terminator]──PC1──X PC2──Teacher PC──PC3──[Terminator]
Backbone: [Terminator]──PC1──PC2──// BREAK //──Teacher PC──PC3

Figure 2. Fault Isolation in the Proposed Network

Table 3. Fault-Isolation Analysis

Fault	Effect	Corrective Action
One workstation fails	Normally limited to that workstation.	Check the computer, interface, and local connector.
Loose connector	The affected section, or possibly the bus, may lose connectivity.	Reconnect or replace the connector.
Backbone cable breaks	All or part of the laboratory loses communication.	Locate and repair or replace the cable.
Missing/incorrect terminator	Signal reflection causes unreliable communication.	Install the correct 50-ohm terminator.
Unmanaged switch fails	Connected workstations lose access.	Replace the switch.

9. Scalability, Security, Advantages, and Limitations

The design is suitable for a small laboratory with a limited number of computers. A few additional workstations may be added if the backbone has sufficient length and connection points. Expansion beyond the intended classroom size becomes difficult because every device shares the same medium. For a larger laboratory, a switched star topology would provide better performance and fault isolation.

Basic security should be applied to the teacher's shared folders through appropriate permissions and strong passwords. File sharing should be enabled only on the classroom network. If internet access is provided, an upstream router or firewall should filter unwanted traffic. These are optional recommendations, not additional requirements stated in the question.

The main advantages are low initial hardware cost, limited cabling, simple installation, and support for light browsing and file sharing without a dedicated server. The main limitations are collisions, limited scalability, difficult troubleshooting, and the possibility that a backbone fault can disrupt the whole laboratory.

10. Testing and Validation

Table 4. Testing and Validation Plan

Test ID	Test Description	Expected Result	Actual Result	Status
T1	Check backbone continuity.	Cable is continuous and correctly terminated.	[TO BE FILLED]	[TO BE FILLED]
T2	Verify workstation connections.	Every workstation detects the network.	[TO BE FILLED]	[TO BE FILLED]
T3	Test communication between two PCs.	Data is transmitted successfully.	[TO BE FILLED]	[TO BE FILLED]
T4	Test teacher-PC file sharing.	Authorized PCs access shared files.	[TO BE FILLED]	[TO BE FILLED]
T5	Disconnect one workstation.	Effect is limited to the disconnected workstation, where applicable.	[TO BE FILLED]	[TO BE FILLED]

11. Tools and Technologies

Table 5. Tools and Technologies

Tool/Technology	Purpose
RG-58 coaxial cable	Traditional shared bus backbone.
T-connectors and BNC connectors	Attach computers to the coaxial backbone.
50-ohm terminators	Prevent signal reflection.
Cat5e/Cat6 UTP cable	Modern low-cost cabling alternative.
Unmanaged Ethernet switch	Contemporary low-cost connectivity option.
Windows/Linux file sharing	Allows the teacher's PC to share resources.
Ping and cable tester	Test connectivity and cable continuity.

12. Problem-Solving Approach and Design Justification

The solution begins by identifying the classroom's constraints: a small number of users, light traffic, limited budget, and basic file and internet access. A bus-oriented design addresses the budget constraint by allowing multiple computers to share one communication line. The teacher's PC provides shared resources, eliminating the need for a dedicated server.

The design recognizes the trade-off between cost and reliability. Although the shared bus is vulnerable to cable faults and collisions, these limitations are acceptable in a supervised classroom with light usage. If the number of users or traffic volume increases, the network should be migrated to a switched star topology.

13. Final Network Architecture

```
Optional Internet
     |
[Router/Gateway]
     |
[RG-58 Bus]——PC1——PC2——Teacher PC——PC3
[50-ohm terminator]            [50-ohm terminator]

Modern option: PC1/PC2/PC3 ——Cat5e—— [8/16-port Unmanaged Switch] —— Teacher PC
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Figure 3. Final Proposed Network Architecture

14. Conclusion

For a small classroom laboratory with a tight budget, a bus topology provides an economical method of connecting computers through a common communication line. The traditional design uses RG-58 coaxial cable, T-connectors, BNC connectors, and 50-ohm terminators, while a modern alternative uses Cat5e/Cat6 cabling and an inexpensive unmanaged switch. The solution supports light classroom activities and permits the teacher's PC to share resources without a dedicated server. Although collisions, limited scalability, and backbone failure are important disadvantages, the trade-off is reasonable for the supervised environment described in Question 3.

References

[1] SIMATS Engineering, "Assessment Tool 2 — CSA07 Computer Networks: Industry Problem-Solving Task," uploaded assignment PDF, Question 3.